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VENTILATOR ASSEMBLY HAVING A VARIABLE FAN CONTROL FOR AIR FLOW IN AN AIR CONDITIONER

Abstract

An air conditioner unit or ventilator assembly may include a ventilator cabinet, a fan, a capacitor assembly, and a user input. The ventilator cabinet may define a path for a flow of air relative to the air conditioner unit. The fan may be mounted within the ventilator cabinet to urge the flow air. The capacitor assembly may be in electrical communication with the fan to control an electrical current to the fan. The capacitor assembly may be configured to selectively operate across a predetermined capacitance range. The user input may be in operable communication with the capacitor assembly. The user input may be configured to set an operable capacitance of the capacitor assembly within the predetermined capacitance range.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] The present subject matter relates generally to air conditioning appliances, and more particularly to assemblies for providing make-up air to air conditioning appliances.

BACKGROUND OF THE DISCLOSURE

[0002] Air conditioner or air conditioning appliance units are conventionally used to adjust the temperature within structures such as dwellings and office buildings. In particular, one-unit type room air conditioner units, such as single-package vertical units (SPVU), may be used to adjust the temperature in, for example, a single room or group of rooms of a structure. A typical one-unit type air conditioner or air conditioning appliance includes an indoor portion and an outdoor portion. The indoor portion generally communicates (e.g., exchanges air) with the area within a building, and the outdoor portion generally communicates (e.g., exchanges air) with the area outside a building. Accordingly, the air conditioner unit generally extends through, for example, an outer wall of the structure. Generally, a fan may be operable to rotate to motivate air through the indoor portion. Another fan may be operable to rotate to motivate air through the outdoor portion. A sealed cooling system including a compressor is generally housed within the air conditioner unit to treat (e.g., cool or heat) air as it is circulated through the indoor portion of the air conditioner unit. One or more control boards are typically provided to direct the operation of various elements of the particular air conditioner unit.

[0003] Frequently, the indoor space may need to draw in air from the outdoors (i.e., make-up air). For example, if a vent fan is turned on in a bathroom or air is otherwise ejected from the indoor space, fresh air from the outdoors is required. Depending on, for example, the efficiency of the weather stripping around doors and windows, some make-up air could simply be drawn into the indoors by cracks or other openings. If such cracks are not sufficient, the flow of make-up air may be insufficient or too slow. Furthermore, government regulations, such as fire codes may require that cracks or openings be eliminated as much as possible-precluding a sufficient flow of make-up air. Accordingly, an air conditioner unit that can allow for the introduction of make-up air into the indoor space would be useful. Unfortunately, previous attempts to provide such make-up air have unsatisfactory. For example, characteristics about both the air conditioner assembly, including any make-up air assembly, must be set at the factory or assembly stage. Existing systems cannot be modified. Moreover, there can be a high degree of variability in the volume of make-up air that may be appropriate or desired. As a result, a user or installer may be required to expend a large amount of time, energy, or money in order to ensure an air conditioner assembly can receive sufficient (and not excessive or insufficient) make-up air. Additionally or alternatively, make-up air module for supplying make-up air from outdoors often suffer from inefficiencies. For example, the addition of outdoor make-up air adds a large amount of heating or cooling load to the space being conditioned.

[0004] As a result, it would be useful to provide an air conditioner unit or ventilation assembly that includes features for addressing one or more of the above issues.

BRIEF DESCRIPTION OF THE DISCLOSURE

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a ventilator assembly for an air conditioner unit is provided. The ventilator assembly may include a ventilator cabinet, a primary fan, a primary capacitor assembly, and a primary user input. The ventilator cabinet may define a primary path for a flow of make-up air to an indoor portion of the air conditioner unit. The primary fan may be mounted within the ventilator cabinet to urge the flow of make-up air into the indoor portion. The primary capacitor assembly may be in electrical communication with the primary fan to control an electrical current to the primary fan. The primary capacitor assembly may be configured to selectively operate across a predetermined primary capacitance range. The primary user input may be in operable communication with the primary capacitor assembly. The primary user input may be configured to set an operable capacitance of the primary capacitor assembly within the predetermined primary capacitance range.

[0007] In another exemplary aspect of the present disclosure, an air conditioner unit is provided. The air conditioner unit may include a cabinet, a bulkhead, an indoor heat exchanger, an outdoor heat exchanger, and a ventilator assembly. The cabinet may define an indoor inlet and an indoor outlet. The bulkhead may be mounted within the cabinet to define an indoor portion and an outdoor portion. The indoor heat exchanger may be mounted within the cabinet at the indoor portion. The outdoor heat exchanger may be mounted within the cabinet at the outdoor portion. The ventilator assembly may be mounted to the cabinet apart from the indoor and outdoor portions, the ventilator assembly may include a ventilator cabinet, a primary fan, a primary capacitor assembly, and a primary user input. The ventilator cabinet may define a primary path for a flow of make-up air to an indoor portion of the air conditioner unit. The primary fan may be mounted within the ventilator cabinet to urge the flow of make-up air into the indoor portion. The primary capacitor assembly may be in electrical communication with the primary fan to control an electrical current to the primary fan. The primary capacitor assembly may be configured to selectively operate across a predetermined primary capacitance range. The primary user input may be in operable communication with the primary capacitor assembly. The primary user input may be configured to set an operable capacitance of the primary capacitor assembly within the predetermined primary capacitance range.

[0008] In yet another exemplary aspect of the present disclosure, a ventilator assembly for an air conditioner unit is provided. The ventilator assembly may include a ventilator cabinet, an auxiliary fan, an auxiliary capacitor assembly, and an auxiliary user input. The ventilator cabinet may define an exhaust path for a flow of exhaust air from a portion of the air conditioner unit to an outdoor environment. The auxiliary fan may be mounted within the ventilator cabinet to urge the flow of make-up air into the indoor portion. The auxiliary capacitor assembly may be in electrical communication with the auxiliary fan to control an electrical current to the auxiliary fan. The auxiliary capacitor assembly may be configured to selectively operate across a predetermined auxiliary capacitance range. The auxiliary user input may be in operable communication with the auxiliary capacitor assembly. The auxiliary user input may be configured to set an operable capacitance of the auxiliary capacitor assembly within the predetermined auxiliary capacitance range.

[0009] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0011] FIG. **1** provides a perspective view of an air conditioning appliance including an energy

recovery ventilator according to an example embodiment of the present disclosure.
[0012] FIG. **2** provides a front elevation view of the exemplary air conditioning appliance of FIG. **1** according to an example embodiment of the present disclosure.
[0013] FIG. **3** provides a section view of the exemplary air conditioning appliance of FIG. **1** according to an example embodiment of the present disclosure.
[0014] FIG. **4** provides an exploded view of the example energy recovery ventilator of FIG. **1** according to an exemplary embodiment of the present disclosure.
[0015] FIG. **5** provides a perspective view of the example energy recovery ventilator of FIG. **1** according to an exemplary embodiment of the present disclosure.
[0016] FIG. **6** provides a perspective view of the example energy recovery ventilator of FIG. **1** with a front cover removed according to an exemplary embodiment of the present disclosure.
[0017] FIG. **7** provides an elevation view of the front cover of the example energy recovery ventilator of FIG. **1** according to an exemplary embodiment of the present disclosure.
[0018] FIG. **8** provides a side, sectional view of the example air conditioner unit and energy recovery ventilator of FIG. **1** according to an exemplary embodiment of the present disclosure.
[0019] FIG. **9** provides another side, sectional view of the example air conditioner unit and energy recovery ventilator of FIG. **1** according to an exemplary embodiment of the present disclosure.
[0020] FIG. **10** provides a front elevation view of the example energy recovery ventilator of FIG. **1** according to an example embodiment of the present disclosure.
[0021] FIG. **11** provides a simplified perspective view of a portion of a capacitor assembly for a ventilator assembly according to exemplary embodiments of the present disclosure.
[0022] FIGS. **12**A and **12**B provides simplified plan views of a portion of the exemplary capacitor assembly of FIG. **11**, wherein the capacitor assembly is rotated between a minimum and maximum capacitance position, respectively.
[0023] FIG. **13** provides a schematic view of a portion of a ventilator assembly for an air conditioner unit according to exemplary embodiments of the present disclosure.
[0024] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0025] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations.
[0026] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The terms “upstream” and “downstream” refer to the relative flow direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the flow direction from which the fluid flows, and “downstream” refers to the flow direction to which the fluid flows. In addition, here and throughout the specification and claims, range limitations may be combined or interchanged. Such ranges are

identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
[0027] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components or systems. For example, the approximating language may refer to being within a 10 percent margin (i.e., including values within ten percent greater or less than the stated value). In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction (e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, such as, clockwise or counterclockwise, with the vertical direction V).
[0028] Except as explicitly indicated otherwise, recitation of a singular processing element (e.g., “a controller,” “a processor,” “a microprocessor,” etc.) is understood to include more than one processing element. In other words, “a processing element” is generally understood as “one or more processing element.” Furthermore, barring a specific statement to the contrary, any steps or functions recited as being performed by “the processing element” or “said processing element” are generally understood to be capable of being performed by “any one of the one or more processing elements.” Thus, a first step or function performed by “the processing element” may be performed by “any one of the one or more processing elements,” and a second step or function performed by “the processing element” may be performed by “any one of the one or more processing elements and not necessarily by the same one of the one or more processing elements by which the first step or function is performed.” Moreover, it is understood that recitation of “the processing element” or “said processing element” performing a plurality of steps or functions does not require that at least one discrete processing element be capable of performing each one of the plurality of steps or functions.
[0029] Generally, aspects of the present disclosure may provide an assembly or appliance in which the volume of make-up air actively supplied by or urged to a room can be easily varied or adjusted after installation of an air conditioning appliance (e.g., without requiring additional mechanical dampers). Additionally or alternatively, aspects of the present disclosure may permit existing air conditioning appliances to be adjusted or improved (e.g., without requiring extensive disassembly or removal of the appliance from its installed location).
[0030] Turning now to the figures, FIGS. **1** through **3** illustrate an exemplary air conditioning appliance (e.g., air conditioner **100**). Specifically, FIG. **1** provides a perspective view of air conditioner **100**, FIG. **2** provides a front view of air conditioner **100**, and FIG. **3** provides a cross sectional view of air conditioner **100**. As shown, air conditioner **100** may be provided as a one-unit type air conditioner **100**, such as a single-package vertical unit (SPVU). However, it should be appreciated that aspects of the present disclosure may be used with other suitable air conditioning units or air filtering devices, such as a packaged terminal air conditioner unit (PTAC), a split heat pump system, etc.
[0031] Air conditioner **100** includes a package housing or cabinet **102** supporting and defining an indoor portion **104** and an outdoor portion **106**. Generally, air conditioner **100** generally defines a vertical direction V, a lateral direction L, and a transverse direction T. Each direction V, L, T is perpendicular to each other, such that an orthogonal coordinate system is generally defined.
[0032] In some embodiments, cabinet **102** contains various other components of the air conditioner **100**. Cabinet **102** may include, for example, a rear opening **110** (e.g., with or without a grill or grate

thereacross) and a front opening **112** (e.g., with or without a grill or grate thereacross) may be spaced apart from each other along the transverse direction T. The rear opening **110** may be part of the outdoor portion **106**, while the front opening **112** is part of the indoor portion **104**. Components of the outdoor portion **106**, such as an outdoor heat exchanger **120**, outdoor fan **124**, and compressor **126** may be enclosed within cabinet **102** between front opening **112** and rear opening **110**. In certain embodiments, one or more components of outdoor portion **106** are mounted on a base pan **136**, as shown. According to exemplary embodiments, base pan **136** may be received within a drain pan (e.g., for collecting condensation formed during operation).

[0033] During certain operations, air **114** may be drawn to outdoor portion **106** through rear opening **110**. Specifically, an outdoor inlet **128** defined through cabinet **102** may receive outdoor air **114** motivated by outdoor fan **124**. Within cabinet **102**, the received outdoor air **114** may be motivated through or across outdoor fan **124**. Moreover, at least a portion of the outdoor air **114** may be motivated through or across outdoor heat exchanger **120** before exiting the rear opening **110** at an outdoor outlet **130**. It is noted that although outdoor inlet **128** is illustrated as being defined above outdoor outlet **130**, alternative embodiments may reverse this relative orientation (e.g., such that outdoor inlet **128** is defined below outdoor outlet **130**) or provide outdoor inlet **128** beside outdoor outlet **130** in a side-by-side orientation, or another suitable orientation.

[0034] As shown, indoor portion **104** may include an indoor heat exchanger **122**, a blower fan **142**, and a heating unit **132**. These components may, for example, be housed behind the front opening **112**. A bulkhead **134** may generally support or house various other components or portions thereof of the indoor portion **104**, such as the blower fan **142**. Bulkhead **134** may generally separate and define the indoor portion **104** and outdoor portion **106** within cabinet **102**. Additionally, or alternatively, bulkhead **134** or indoor heat exchanger **122** may be mounted on base pan **136** (e.g., at a higher vertical position than outdoor heat exchanger **120**), as shown.

[0035] During certain operations, air **116** may be drawn to indoor portion **104** through front opening **112**. Specifically, an indoor inlet **138** defined through cabinet **102** may receive indoor air **116** motivated by blower fan **142**. At least a portion of the indoor air **116** may be motivated through or across indoor heat exchanger **122** (e.g., before passing to bulkhead **134**). From blower fan **142**, indoor air **116** may be motivated (e.g., across heating unit **132**) and returned to the indoor area of the room through an indoor outlet **140** defined through cabinet **102** (e.g., above indoor inlet **138** along the vertical direction V). Optionally, one or more conduits (not pictured) may be mounted on or downstream from indoor outlet **140** to further guide air **116** from air conditioner **100**. It is noted that although indoor outlet **140** is illustrated as generally directing air upward, it is understood that indoor outlet **140** may be defined in alternative embodiments to direct air in any other suitable direction.

[0036] Outdoor and indoor heat exchanger **120**, **122** may be components of a thermodynamic assembly (i.e., sealed system), which may be operated as a refrigeration assembly (and thus perform a refrigeration cycle) or, in the case of the heat pump unit embodiment, a heat pump (and thus perform a heat pump cycle). Thus, as is understood, exemplary heat pump unit embodiments may be selectively operated perform a refrigeration cycle at certain instances (e.g., while in a cooling mode) and a heat pump cycle at other instances (e.g., while in a heating mode). By contrast, exemplary A/C exclusive unit embodiments may be unable to perform a heat pump cycle (e.g., while in the heating mode), but still perform a refrigeration cycle (e.g., while in a cooling mode).

[0037] The sealed system may, for example, further include compressor **126** (e.g., mounted on base pan **136**) and an expansion device (e.g., expansion valve or capillary tube—not pictured), both of which may be in fluid communication with the heat exchangers **120**, **122** to flow refrigerant therethrough, as is generally understood. The outdoor and indoor heat exchanger **120**, **122** may each include coils **146**, **148**, as illustrated, through which a refrigerant may flow for heat exchange purposes, as is generally understood.

[0038] According to an example embodiment, compressor **126** may be a variable speed compressor. In this regard, compressor **126** may be operated at various speeds depending on the current air conditioning needs of the room and the demand on the sealed system. For example, according to an exemplary embodiment, compressor **126** may be configured to operate at any speed between a minimum speed [e.g., 1500 revolutions per minute (RPM)] to a maximum rated speed (e.g., 3500 RPM). Notably, the use of variable speed compressor **126** enables efficient operation of the sealed system, minimizes unnecessary noise when compressor **126** does not need to operate at full speed, and ensures a comfortable environment within the room.
[0039] According to exemplary embodiments, air conditioner **100** may further include a plenum **144** to direct air to or from cabinet **102**. When installed, plenum **144** may be selectively attached to (e.g., fixed to or mounted against) cabinet **102** (e.g., via a suitable mechanical fastener, adhesive, gasket, etc.) and extend through a structure wall **150** (e.g., an outer wall of the structure within which air conditioner **100** is installed) and above a floor of the structure. In particular, plenum **144** extends along an axial direction X (e.g., parallel to the transverse direction T) through a hole or channel **152** in the structure wall **150** that passes from an internal surface **154** to an external surface **156**. In addition, it should be appreciated that plenum **144** may be formed from two or more telescoping structures (e.g., to accommodate different thicknesses of structure wall **150**).
[0040] The operation of air conditioner **100** including compressor **126** (and thus the sealed system generally), blower fan **142**, outdoor fan **124**, heating unit **132**, and other suitable components may be controlled by a control board or controller **158**. Controller **158** may be in communication (via for example a suitable wired or wireless connection) to such components of the air conditioner **100**. By way of example, the controller **158** may include a memory and one or more processing devices such as microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of air conditioner **100**. The memory may be a separate component from the processor or may be included onboard within the processor. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH.
[0041] Air conditioner **100** may additionally include a control panel **160** and one or more user inputs **162**, which may be included in control panel **160**. The user inputs **162** may be in communication with the controller **158**. A user of the air conditioner **100** may interact with the user inputs **162** to operate the air conditioner **100**, and user commands may be transmitted between the user inputs **162** and controller **158** to facilitate operation of the air conditioner **100** based on such user commands. A display **164** may additionally be provided in control panel **160** and may be in communication with the controller **158**. Display **164** may, for example be a touchscreen or other text-readable display screen, or alternatively may simply be a light that can be activated and deactivated as required to provide an indication of, for example, an event or setting for the air conditioner **100**.
[0042] As explained briefly above, it may be desirable to periodically supplement the indoor air with make-up air from the outdoors. For example, in some cases it may be desirable to allow outside air (i.e., “make-up air”) to flow into the room (e.g., in order to meet government regulations, to compensate for negative pressure created within the room, etc.). In addition, it may be desirable to treat or condition make-up air prior to blowing it into the room (e.g., when there is a relatively large difference between the outdoor air temperature or humidity relative to target indoor temperature or humidity levels). For example, if it is very cold outside, it may be desirable to heat the flow of make-up air prior to passing it into the room (e.g., for improved occupant comfort and improved system efficiency). Accordingly, aspects of the present disclosure are generally directed to features of air conditioner **100** that may facilitate the supply of conditioned make-up air into the room.
[0043] Referring now generally to FIGS. **1** through **10**, air conditioner **100** may generally include an ventilator assembly **200** that is generally configured for supplying a flow of make-up air (e.g.,

identified herein generally by reference numeral **202**) into indoor portion **104** (e.g., for mixing with a flow of indoor air **116**). In optional embodiments, ventilator assembly **200** may be configured for collecting a flow of exhaust air (e.g., identified herein generally by reference numeral **204**) and transferring thermal energy between the flow of make-up air **202** and the flow of exhaust air **204**. Ventilator assembly **200** will be described in more detail below according to an example embodiment. However, it should be appreciated that the embodiment described as only exemplary and is not intended to limit the scope of the present disclosure in any manner.

[0044] According to the illustrated embodiment, ventilator assembly **200** may generally be mounted or fixed to cabinet **102** of air conditioner **100**. Specifically, according to the illustrated embodiment, ventilator assembly **200** is positioned on top of cabinet **102**, though other suitable positions are possible and within the scope of the present disclosure. According to an example embodiment, ventilator assembly **200** generally includes a ventilator cabinet **210** through which the flow of make-up air **202** and the flow of exhaust air **204** are passed. Specifically, ventilator cabinet **210** may generally define a primary flow path **212** through which the flow of make-up air **202** passes and an energy recovery or exhaust path **214** through which the flow of exhaust air **204** passes.

[0045] Although ventilator assembly **200** and ventilator cabinet **210** are described herein as being used with air conditioner unit **100** and being mounted directly thereon, it should be appreciated that ventilator assembly **200** and ventilator cabinet **210** may be used with any suitable air conditioner/heat pump system. Furthermore, any suitable positioning of the ventilator cabinet **210** is possible (e.g., remote from air conditioner unit) and any suitable routing or ducting of the primary flow path **212** and the energy recovery path **214** are possible and within the scope of the present disclosure.

[0046] Specifically, according to the illustrated embodiment, ventilator cabinet **210** may generally include a rear wall **220**, the front wall **222**, and a plurality of sidewalls **224** that generally define a square or rectangular box that defines a ventilator plenum **226**. Notably, according to an example embodiment, in order to retain thermal energy within the ventilator cabinet **210**, ventilator cabinet **210** may generally be an insulated housing (e.g., such that the interior surfaces of ventilator cabinet **210** may be covered in an insulating material, may be formed from insulative layers, or may otherwise be suitably thermally insulated).

[0047] Referring now specifically to FIGS. **4** through **10**, ventilator cabinet **210** may generally define a fresh air inlet **230** that is fluidly coupled to an ambient environment (e.g., the outdoors). In this manner, air conditioner **100** may draw in the flow of make-up air **202** from outside of the building where air conditioner **100** is located and may pass the flow of make-up air **202** through ventilator plenum **226**. In addition, ventilator cabinet **210** may define a fresh air outlet **232** that is fluidly coupled to the indoor portion **104** of air conditioner **100**. As best shown in FIGS. **8** through **10**, fresh air inlet **230** and fresh air outlet **232** generally form a portion of primary flow path **212**.

[0048] In addition, ventilator cabinet **210** may generally define an exhaust inlet **234** that is fluidly coupled to an interior space where air conditioner **100** is located. In this regard, for example, exhaust inlet **234** may be fluidly coupled to a return air duct system from an indoor space that is conditioned by air conditioner **100**. For example, this return air duct system may collect indoor air from a connected bathroom, multiple rooms of a suite, or other areas within the indoor space. Ventilator cabinet **210** may further define an exhaust outlet **236** that is fluidly coupled to the ambient environment (e.g., the outdoors). Exhaust inlet **234** and exhaust outlet **236** may generally form a portion of energy recovery path **214**.

[0049] Notably, as described briefly above, the flow of make-up air **202** from outdoor portion **106** may have a temperature that would be uncomfortable if directed into the indoor portion **104** without being conditioned. For example, if it is very cold outside, the flow of make-up air **202** may be cool or frigid, and injecting such air into the flow of indoor air **116** may result in poor system efficiency and user dissatisfaction. By contrast, if it is very warm outside, it may be desirable to

lower the temperature of the flow of make-up air **202** before injecting it into the room.
[0050] Accordingly, ventilator assembly **200** may further include a heat exchanger **240** that is positioned within ventilator plenum **226** for transferring thermal energy to or from the flow of make-up air **202** such that the temperature of the flow of make-up air **202** is more desirable and appropriate to the conditioning needs of the room. Specifically, according to the illustrated embodiment, heat exchanger **240** may generally define a portion of primary flow path **212** through which the flow of make-up air **202** passes into the room. In addition, heat exchanger **240** may define a portion of energy recovery path **214**. As explained herein, energy recovery path **214** may generally be configured for receiving the flow of exhaust air **204** that is configured for transferring heat with the flow of make-up air **202** (e.g., via heat exchanger **240**).
[0051] More specifically, according to example embodiments, primary flow path **212** and energy recovery path **214** are fluidly isolated from each other while being thermally coupled to each other. For example, according to the illustrated embodiment, heat exchanger **240** is a crossflow air-to-air heat exchanger. In this regard, the flow of make-up air **202** and the flow of exhaust air **204** may flow through fluidly isolated passages perpendicular to each other such that thermal energy may pass between the two flows while they may remain fluidly isolated. For example, according to the illustrated embodiment, heat exchanger **240** comprises a plurality of thin plates stacked along the transverse direction T. These plates at least partially define primary flow path **212** and energy recovery path **214** and may be used to separate the two flows.
[0052] Optionally, ventilator assembly **200** may generally include a plurality of retainer brackets that are positioned within ventilator plenum **226** and generally define a boundary of a heat exchanger cavity **244**. For example, according to the illustrated embodiment, ventilator assembly **200** includes four retainer brackets **242** that are positioned on the four sidewalls **224** of ventilator cabinet **210**. These retainer brackets **242** generally extend from sidewalls **224** into ventilator plenum **226** to sealingly engage respective corners of heat exchanger **240**. More specifically, a distal end **246** of each of retainer brackets **242** may generally define a notch or V-shaped slot for receiving heat exchanger **240**. In addition, it should be appreciated that distal ends **246** may include a sealing material or gasket (not shown) for preventing air leakage between retainer brackets **242** and heat exchanger **240**.
[0053] In addition, ventilator assembly **200** may include a plurality of mounting gaskets **250** that are positioned between retainer brackets **242** and ventilator cabinet **210** to provide a fluid seal and to divide ventilator plenum **226** into four quadrants that are at least partially fluidly isolated from each other (e.g., thereby defining primary flow path **212** and energy recovery path **214**). In operation, the flow of make-up air **202** may enter fresh air inlet **230** and may pass through heat exchanger **240** before exiting fresh air outlet **232** along the primary flow path **212**. Similarly, the flow of exhaust air **204** may enter exhaust inlet **234** and may pass through heat exchanger **240** before exiting exhaust outlet **236** along the energy recovery path **214**. Notably, the flow of make-up air **202** and the flow of exhaust air **204** may transfer thermal energy between each other in heat exchanger **240** while remaining fluidly isolated.
[0054] According to the illustrated embodiments, ventilator assembly **200** includes one or more fan assemblies for selectively urging the flow of make-up air **202** or the flow of exhaust air **204**. In this regard, as illustrated, ventilator assembly **200** may generally include a primary fan **260** fluidly coupled to primary flow path **212** for urging the flow of make-up air **202** into indoor portion **104** (e.g., via fresh air outlet **232**). As shown, primary fan **260** may be positioned on a downstream end of heat exchanger **240**. In addition, ventilator assembly **200** may include an auxiliary fan **262** that is fluidly coupled to energy recovery path **214** for drawing a flow of exhaust air **204** through energy recovery path **214**. According to the illustrated embodiment, auxiliary fan **262** is positioned on the downstream end of heat exchanger **240**. According to the illustrated embodiment, primary fan **260** and auxiliary fan **262** are positioned within ventilator cabinet **210**, though other suitable positions are possible and within the scope of the present disclosure.

[0055] According to the illustrated embodiment, both primary fan **260** and auxiliary fan **262** are centrifugal fans. However, it should be appreciated that any other suitable number, type, and configuration of fan or blower could be used to urge a flow of make-up air **202** and the flow of exhaust air **204** according to alternative embodiments. In addition, primary fan **260** and auxiliary fan **262** may be positioned in any other suitable location within ventilator assembly **200**. The embodiments described herein are only exemplary and are not intended to limit the scope of the present disclosure.
[0056] As best shown in FIGS. **1**, **2**, **8**, and **9**, air-conditioner **100** may further define a standoff plenum **270** that at least partially define indoor inlet **138** of cabinet **102**. In this regard, standoff plenum **270** may generally be positioned upstream of indoor heat exchanger **122** and may be configured to receive the flow of indoor air **116** and the flow of make-up air **202**. In addition, air-conditioner **100** may further include a make-up air duct **272** that provides fluid communication with fresh air outlet **232** and a make-up air supply port **274** that is defined on standoff plenum **270**. In this manner, the conditioned flow of make-up air **202** may exit ventilator assembly **200**, pass through make-up air duct **272**, and interior into standoff plenum **270** where it mixes with the flow of indoor air **116**.
[0057] Generally, a power source **252** is operably (e.g., electrically) coupled to primary fan **260** or auxiliary fan **262** (e.g., through an electrical receptacle or other connection to the ventilator assembly **200**) for initiating the flow of make-up air **202** or the flow of exhaust air **204**, respectively. In particular, the power source **252** may supply an electrical current (e.g., directly, such as independently of controller **158**, or indirectly, such as via controller **158**) to primary fan **260** or auxiliary fan **262**. It should be appreciated that these fans **260**, **262** may each be operated at the same speed or at a different speed. For example, the flow rate of the make-up air **202** is typically dictated by the make-up air needs within the room (e.g., based on government regulations, occupancy, etc.). Similarly, the outside temperature and humidity may require different flow rates of exhaust air **204** to suitably condition the flow of make-up air **202** prior to introduction into the corresponding room.
[0058] In some embodiments, the flow of make-up air **202** or the flow of exhaust air **204** can be selectively varied, such as by adjustments made to rotation the speed of primary fan **260** or auxiliary fan **262**, respectively. In particular, such adjustments may be made independently of the controller **158**. For instance, turning briefly to FIG. **13**, a schematic view of ventilator assembly **200** is provided, illustrating an electrical connection between the power source **252** and the primary fan **260**, as well as the power source **252** and the primary fan **260**. As shown, one or more variable capacitor assemblies **256**, **258** may be in electrical communication with the fans **260**, **262**, such as to vary the capacitance of the circuit to one or more of the fans **260**, **262** from the power source **252**.
[0059] As an example, a primary capacitor assembly **256** may be included in electrical communication with the primary fan **260** (e.g., exclusively or, alternatively, in conjunction with electrical communication with another fan, such as auxiliary fan **262**). When assembled, primary capacitor assembly **256** may generally control an electrical current to the primary fan **260** (or auxiliary fan **262**) from the power source **252**. Specifically, the primary capacitor assembly **256** may be configured to selectively operate across a predetermined primary capacitance range (e.g., 1 to 7 microfarads). In other words, the primary capacitor assembly **256** may be set to include an infinite or discrete number of user-selectable operable capacitance settings between the predetermined primary capacitance range. Thus, the primary capacitor assembly **256** may be able to operate at the minimum capacitance of the capacitance range, the maximum capacitance of the capacitance range, or one or more intermediate capacitances within the capacitance range. Depending on the set or operable capacitance of the primary capacitor assembly **256**, the rotational speed of the primary fan **260** (or auxiliary fan **262**)—and thereby the target flow rate for the flow of make-up air **202** (or exhaust air **204**)—may be varied. For instance, the voltage delivered to the

primary fan **260** or auxiliary fan **262** (e.g., such that a single capacitor assembly controls the electrical current to both primary fan **260** and **262** simultaneously) may be varied according to the operable capacitance, which in turn, may vary the rotational speed of the primary fan **260** or exhaust air **204**, as would be understood.

[0060] As an additional or alternative example, an auxiliary capacitor assembly **258** may be included in electrical communication with the auxiliary fan **262**. When assembled, auxiliary capacitor assembly **258** may generally control an electrical current to the auxiliary fan **262** from the power source **252**. Specifically, the auxiliary capacitor assembly **258** may be configured to selectively operate across a predetermined auxiliary capacitance range (e.g., 1 to 7 microfarads). In other words, the auxiliary capacitor assembly **258** may be set to include an infinite or discrete number of user-selectable operable capacitance settings between the predetermined auxiliary capacitance range. Thus, the auxiliary capacitor assembly **258** may be able to operate at the minimum capacitance of the capacitance range, the maximum capacitance of the capacitance range, or one or more intermediate capacitances within the capacitance range. Depending on the set or operable capacitance of the auxiliary capacitor assembly **258**, the rotational speed of the auxiliary fan **262**—and thereby the target flow rate for the flow of exhaust air **204**—may be varied. For instance, the voltage of the current delivered to the auxiliary fan **262** may be varied according to the operable capacitance, which in turn, may vary the rotational speed of the auxiliary fan **262**, as would be understood.

[0061] In some embodiments, a primary user input **264** is in operable (e.g., mechanical or electrical) communication with the primary capacitor assembly **256** to set the operable or operating capacitance of the primary capacitor assembly **256**. Specifically, the primary user input **264** may be configured to set the operable capacitance of the primary capacitor assembly **256** (e.g., at a selected level or capacitance within the predetermined primary capacitance range). As shown in FIGS. **1**, **2**, and **10**, the primary user input **264** may be mounted on the ventilator cabinet **210**. Notably a user or installer may have reliable or easy access to the primary user input **264**, such as to vary rotational speed or target flow rate of make-up air **202** without requiring a separate mechanical damper or access to an interface for the controller **158** (or programming for the same). In general, the target flow rate for the flow of make-up air **202** may be between about 10 and 60 cubic feet per minute (cfm), between about 20 and 50 cfm, between about 30 and 40 cfm, or about 35 cfm.

[0062] In additional or alternative embodiments, an auxiliary user input **266** is in operable (e.g., mechanical or electrical) communication with the auxiliary capacitor assembly **258** to set the operable or operating capacitance of the auxiliary capacitor assembly **258** (e.g., apart from or independently of the primary capacitor assembly **256**). Specifically, the auxiliary user input **266** may be configured to set the operable capacitance of the auxiliary capacitor assembly **258** (e.g., at a selected level or capacitance within the predetermined primary auxiliary range). As shown in FIGS. **1**, **2**, and **10**, the auxiliary user input **266** may be mounted on the ventilator cabinet **210** (e.g., spaced apart from the primary user input **264**). Notably a user or installer may have reliable or easy access to the auxiliary user input **266**, such as to vary rotational speed or target flow rate of exhaust air **204** without requiring a separate mechanical damper or access to an interface for the controller **158** (or programming for the same). In general, the target flow rate for the flow of exhaust air **204** may be between about 10 and 60 cubic feet per minute (cfm), between about 20 and 50 cfm, between about 30 and 40 cfm, or about 35 cfm.

[0063] It should be appreciated that primary fan **260**, auxiliary fan **262**, and blower fan **142** may be collectively operated in order to ensure that the flow of make-up air **202** is provided at a target flow rate while the outlet temperature of indoor air **116** remains within a desired range. For example, according to an example embodiment, these fans are operating to mix one part of the flow of make-up air **202** to every three parts of the flow of indoor air **116**. Other suitable mix ratios are possible and within the scope of the present disclosure.

[0064] Generally, the primary capacitor assembly **256** or the auxiliary capacitor assembly **258** may be provided with or as any suitable arrangement of capacitors. For instance, the primary capacitor assembly **256** or the auxiliary capacitor assembly **258** may include a plurality of selectable fixed-capacitance capacitors (e.g., wired in electrical communication with each other). In other words, multiple discrete capacitors having a fixed capacitance rating may be wired in the same circuit (e.g., in parallel). In some such embodiments, the corresponding user input may include a plurality of discrete switches each coupled to at least one capacitor such that individual capacitors may be bypassed or "turned off" to vary the operable capacitance of the overall primary capacitor assembly **256** or auxiliary capacitor assembly **258**, as would be understood in light of the present disclosure.

[0065] Turning especially to FIGS. **11**, **12**, and **12**B, the primary capacitor assembly **256** or the auxiliary capacitor assembly **258** may further be provided as or include a variable run capacitor (e.g., separate from or in addition to any fixed-capacitance capacitors). Such a variable run capacitor may be configured to vary operable capacitance, for instance, based on the adjustable physical relationship between two or more corresponding elements. As an example, a rotor **268**A and stator **268**B may be provided. When assembled, the rotor **268**A may be rotated relative to the stator **268**B, such as to vary the operable capacitance at the variable run capacitor between a minimum capacitance (e.g., FIG. **12**A in which the rotor **268**A is held at a maximum preset distance from the stator **268**B) and a maximum capacitance (e.g., FIG. **12**B in which the rotor **268**A is a held at a minimum preset distance from the stator **268**B), as would be understood. In some such embodiments, the corresponding user input **264** or **266** may thus include a knob in mechanical communication with the rotor **268**A (e.g., fixed within or with respect to each other) to rotate the rotor **268**A relative to the stator **268**B. In turn, rotation of the knob may be translated or transferred to the rotor **268**A such rotation of one affects the other, as would further be understood.

[0066] Returning generally to FIGS. **1** through **13**, it may be desirable to periodically stop the flow of make-up air **202** or the flow of exhaust air **204**. Accordingly, air conditioner **100** or ventilator assembly **200** may further include one or more doors that are pivotable between an open and closed position to permit or restrict the respective flows of air. For example, vent doors (not shown) may be positioned in the ventilator assembly **200** or any other suitable location within air conditioner **100** for selectively restricting the flow of make-up air **202** or the flow of exhaust air **204**.

[0067] Additionally or alternatively, it may be desirable to filter the flow of make-up air **202** before introducing it into indoor portion **104**. Accordingly, air conditioner **100** may further include a filter **276** that is positioned within primary flow path **212** for filtering the flow of make-up air **202**. It should be appreciated that filter **276** may be any suitable type, position, and configuration of filtering mechanisms. For example, filter **276** may be a screen filter, a pleated filter, an electrostatic filter, a carbon filter, a fiber glass filter, or any other suitable type of filter. Although filter **276** is illustrated as being positioned within make-up air duct **272** (e.g., at make-up air supply port **274**) it should be appreciated that filter **276** may alternatively be positioned at any other suitable location within primary flow path **212**.

[0068] Notably, it may be desirable to have access to heat exchanger **240** (e.g., for periodic maintenance, to clear clogs, or to otherwise improve the operating efficiency of ventilator assembly **200**). Accordingly, ventilator cabinet **210** may generally define an access opening **280** for providing access to heat exchanger cavity **244**. Notably, it is desirable to maintain a good fluid seal between ventilator cabinet **210** and heat exchanger **240** (e.g., to ensure a fluid isolation between primary flow path **212** and energy recovery path **214**). Accordingly, heat exchanger **240** may define a tight or snug fit within heat exchanger cavity **244**.

[0069] Specifically, according to an example embodiment, mounting gaskets **250** may be compressible such that retainer brackets **242** may move or flex slightly relative to sidewalls **224** of ventilator cabinet **210**. In this manner, as a user is inserting heat exchanger **240** into heat exchanger cavity **244**, retainer brackets **242** may be displaced to insert heat exchanger **240** while the resilient nature of mounting gaskets **250** rebound to form a strong seal with heat exchanger **240** when the retainer brackets **242** are no longer manually displaced.

[0070] As best shown in FIGS. **4** and **5**, ventilator assembly **200** may further include an access cover **282** that is removably mounted over access opening **280** for providing selective access to heat exchanger cavity **244**. Notably, due to the limited space available in locations where air conditioner **100** is typically installed, access cover **282** may be fully removable instead of being a hinged cover. It should be appreciated that a gasket (not shown) may be positioned or defined on an inner face of access cover **282** (e.g., to form a fluid seal with a front **284** of heat exchanger **240**). In addition, according to the illustrated embodiment, heat exchanger **240** includes a handle **286** that extends from front **284** and access cover **282** finds an aperture **288** for receiving the handle **286**.
[0071] According to the illustrated embodiment, in order to secure access cover **282** over access opening **280**, ventilator assembly **200** may further include a plurality of rotatable clips **290** that are positioned on an outside of ventilator cabinet **210**. More specifically, according to an example embodiment, rotatable clips **290** may be rotatable and slidable relative to front wall **222** of ventilator cabinet **210**. In this manner, rotatable clips **290** may be fully recessed or moved out of the way relative to access opening **280** (e.g., to facilitate the insertion of heat exchanger **240** into a heat exchanger cavity **244**). After heat exchanger **240** is properly installed, rotatable clips **290** may be rotated and slid to secure access cover **282** against front wall **222** of ventilator cabinet **210** and front **284** of heat exchanger **240**.
[0072] Specifically, according to the illustrated embodiment, rotatable clips **290** are mounted directly to retainer brackets **242** such that movement of retainer brackets **242** (e.g., through compression of mounting gaskets **250**) permit the rotatable clips **290** to be retracted relative to access opening **280** while springing back toward access opening **280** when released after heat exchanger **240** is installed. In this manner, rotatable clips **290** securely and sealingly engage access cover **282** over access opening **280**. For example, the plurality of retainer brackets **242** and the plurality of rotatable clips **290** may be configured to slide between about 1/32 and ½ of an inch, between about 1/16 and ¼ of an inch, or about ⅛ of an inch.
[0073] Notably, in order to permit the sliding of rotatable clips **290**, front wall **222** of ventilator cabinet **210** may generally define an elongated slot **292** (e.g., as shown in FIG. **7**). In addition, retainer brackets **242** may each define a stud **294** is configured for engaging a mounting aperture **296** of each respective rotatable clip **290**, thereby rotatably coupling rotatable clips **290** to retainer brackets **242**. As shown, elongated slots **292** permit the movement of retainer brackets **242** and rotatable clips **290** during installation of heat exchanger **240** (e.g., relative to fixed front wall **222** of ventilator cabinet **210**). Although four retainer brackets **242** and rotatable clips **290** are illustrated, it should be appreciated that ventilator assembly **200** may include any other suitable number, type, position, and configuration of retaining brackets and rotatable clips.
[0074] As explained herein, aspects of the present disclosure are generally directed to a method of mixing fresh air (e.g., make-up air) in a vertical air conditioner. A ventilator kit with an air-to-air heat exchanger may be located atop the vertical air conditioner. Fresh air may be introduced via a duct into the ventilator kit on the lower right-hand side where air is then sucked into a cross flow plate exchanger with the unconditioned exhaust stream of the building moving perpendicular to the fresh air intake. Once passing through the crossflow plate (air to air) heat exchanger and exchanging sensible and latent energy with the exhaust stream, the conditioned fresh air stream may be collected in an insulated cavity on the upper left-hand side of the ventilator. A centrifugal fan may induce a vacuum within the upper left hand side cavity drawing air from the plate heat exchanger and pushing it out the front of the ventilator. A duct may then be connected between the upper left hand side cavity to the front stand-off plenum of the vertical air conditioner. The standoff plenum may direct the conditioned fresh air in front of the room recirculation path of the vertical air conditioner such that room air is mixed with the conditioned fresh air being provided by the ventilator.
[0075] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and

using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A ventilator assembly for an air conditioner unit, the ventilator assembly comprising: a ventilator cabinet defining a primary path for a flow of make-up air to an indoor portion of the air conditioner unit; a primary fan mounted within the ventilator cabinet to urge the flow of make-up air into the indoor portion; a primary capacitor assembly in electrical communication with the primary fan to control an electrical current to the primary fan, the primary capacitor assembly being configured to selectively operate across a predetermined primary capacitance range; and a primary user input in operable communication with the primary capacitor assembly, the primary user input being configured to set an operable capacitance of the primary capacitor assembly within the predetermined primary capacitance range.
2. The ventilator assembly of claim 1, wherein the primary capacitor assembly comprises a plurality of selectable fixed-capacitance capacitors.
3. The ventilator assembly of claim 1, wherein the primary capacitor assembly comprises a variable run capacitor.
4. The ventilator assembly of claim 3, wherein the variable run capacitor comprises a rotor and stator, and wherein the primary user input comprises a knob in mechanical communication with the rotor to rotate the rotor relative to the stator.
5. The ventilator assembly of claim 1, wherein the primary user input is mounted on the ventilator cabinet.
6. The ventilator assembly of claim 1, wherein the ventilator cabinet further defines an energy recovery path for a flow of exhaust air from the air conditioner unit, and wherein the ventilator assembly further comprises: a heat exchanger positioned within the ventilator cabinet along the energy recovery path; and an auxiliary fan mounted within the ventilator cabinet to urge exhaust air through the energy recovery path.
7. The ventilator assembly of claim 6, wherein the ventilator cabinet is attached to an air plenum receivable within a structure wall, the ventilator cabinet defining a fresh air inlet downstream from the air plenum and a fresh air outlet above and downstream from the fresh air inlet along the primary path, the ventilator cabinet further defining an exhaust inlet and an exhaust outlet along the energy recovery path.
8. The ventilator assembly of claim 6, further comprising: an auxiliary capacitor assembly in electrical communication with the auxiliary fan to control an electrical current to the auxiliary fan, the auxiliary capacitor assembly being configured to selectively operate across a predetermined auxiliary capacitance range.
9. The ventilator assembly of claim 8, further comprising: an auxiliary user input spaced apart from the primary user input and in operable communication with the auxiliary capacitor assembly, the auxiliary user input being configured to set an operable capacitance of the auxiliary capacitor assembly within the predetermined auxiliary capacitance range.
10. An air conditioner unit defining a vertical, a lateral, and a transverse direction, the air conditioner unit comprising: a cabinet defining an indoor inlet and an indoor outlet; a bulkhead mounted within the cabinet to define an indoor portion and an outdoor portion; an indoor heat exchanger mounted within the cabinet at the indoor portion; an outdoor heat exchanger mounted within the cabinet at the outdoor portion; and a ventilator assembly mounted to the cabinet apart

from the indoor and outdoor portions, the ventilator assembly comprising: a ventilator cabinet defining a primary path for a flow of make-up air to the indoor portion of the air conditioner unit, a primary fan mounted within the ventilator cabinet to urge the flow of make-up air into the indoor portion, a primary capacitor assembly in electrical communication with the primary fan to control an electrical current to the primary fan, the primary capacitor assembly being configured to selectively operate across a predetermined primary capacitance range, and a primary user input in operable communication with the primary capacitor assembly, the primary user input being configured to set an operable capacitance of the primary capacitor assembly within the predetermined primary capacitance range.

11. The air conditioner unit of claim 10, wherein the primary capacitor assembly comprises a plurality of selectable fixed-capacitance capacitors.

12. The air conditioner unit of claim 10, wherein the primary capacitor assembly comprises a variable run capacitor.

13. The air conditioner unit of claim 12, wherein the variable run capacitor comprises a rotor and stator, and wherein the primary user input comprises a knob in mechanical communication with the rotor to rotate the rotor relative to the stator.

14. The air conditioner unit of claim 10, wherein the primary user input is mounted on the ventilator cabinet.

15. The air conditioner unit of claim 10, wherein the ventilator cabinet further defines an energy recovery path for a flow of exhaust air from the air conditioner unit, and wherein the ventilator assembly further comprises: a heat exchanger positioned within the ventilator cabinet along the energy recovery path; and an auxiliary fan mounted within the ventilator cabinet to urge exhaust air through the energy recovery path.

16. The air conditioner unit of claim 15, wherein the ventilator cabinet is attached to an air plenum receivable within a structure wall, the ventilator cabinet defining a fresh air inlet downstream from the air plenum and a fresh air outlet above and downstream from the fresh air inlet along the primary path, the ventilator cabinet further defining an exhaust inlet and an exhaust outlet along the energy recovery path.

17. The air conditioner unit of claim 15, wherein the ventilator assembly further comprises: an auxiliary capacitor assembly in electrical communication with the auxiliary fan to control an electrical current to the auxiliary fan, the auxiliary capacitor assembly being configured to selectively operate across a predetermined auxiliary capacitance range.

18. The air conditioner unit of claim 17, wherein the ventilator assembly further comprises: an auxiliary user input spaced apart from the primary user input and in operable communication with the auxiliary capacitor assembly, the auxiliary user input being configured to set an operable capacitance of the auxiliary capacitor assembly within the predetermined auxiliary capacitance range.

19. A ventilator assembly for an air conditioner unit, the ventilator assembly comprising: a ventilator cabinet defining an exhaust path for a flow of exhaust air from a portion of the air conditioner unit to an outdoor environment; an auxiliary fan mounted within the ventilator cabinet to urge the flow of exhaust air to the outdoor environment; an auxiliary capacitor assembly in electrical communication with the auxiliary fan to control an electrical current to the auxiliary fan, the auxiliary capacitor assembly being configured to selectively operate across a predetermined auxiliary capacitance range; and an auxiliary user input in operable communication with the auxiliary capacitor assembly, the auxiliary user input being configured to set an operable capacitance of the auxiliary capacitor assembly within the predetermined auxiliary capacitance range.